2024 AMC 10A Problem 10 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 10. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 10

Problem:

Consider the following operation. Given a positive integer n, if n is a multiple of 3 , then you replace n by $\frac{n}{3}$. If n is not a multiple of 3 , then you replace n by $n+10$. Then continue this process. For example, beginning with $n=4$, this procedure gives $4 \rightarrow 14 \rightarrow 24 \rightarrow 8 \rightarrow 18 \rightarrow 6 \rightarrow 2 \rightarrow 12 \rightarrow \cdots$. Suppose you start with $n=100$. What value results if you perform this operation exactly 100 times?

Answer Choices:

A. 10

B. 20

C. 30

D. 40

E. 50

Solution:

The first several iterations give

$$100 \rightarrow 110 \rightarrow 120 \rightarrow 40 \rightarrow 50 \rightarrow 60 \rightarrow 20 \rightarrow 30 \rightarrow 10 \rightarrow 20 \rightarrow 30 \rightarrow 10 \rightarrow \cdots$$

The values will then continue to cycle through $20 \to 30 \to 10$. Note that the value 20 occurs after $6, 9, 12, 15, \ldots$ operations; the value 30 occurs after $7, 10, 13, 16, \ldots$ operations; and the value 10 occurs after $8, 11, 14, 17, \ldots$ operations. Because 100 has remainder 1 when divided by 3 , the value after 100 operations is $(\mathrm{c})\boxed{30}$.